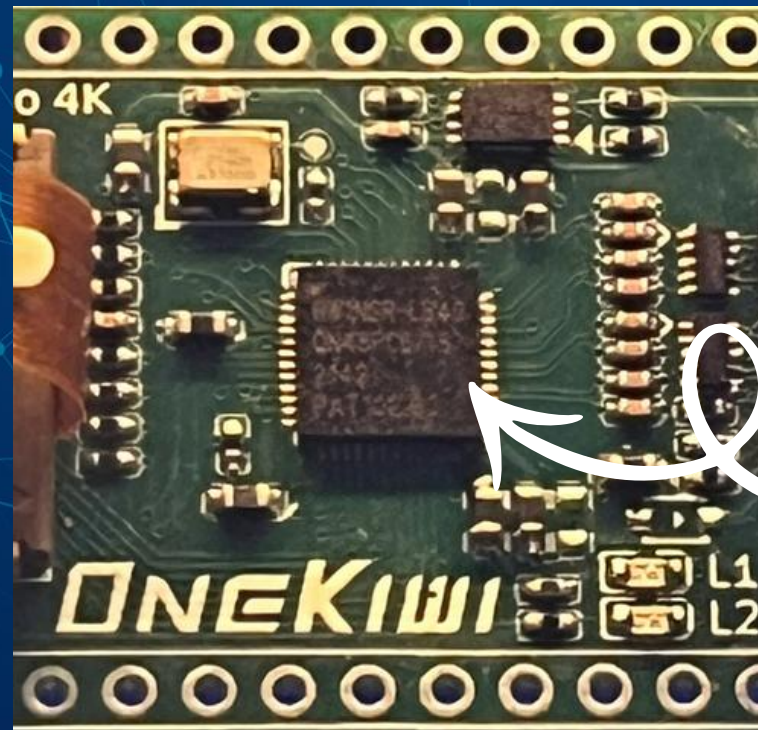


DESIGN OF A REAL-TIME CAMERA NOISE REDUCTION SYSTEM ON FPGA

HD 720p @ 60FPS | Zero Latency



CHIP(芯片): GW1NSR-LV4CQN48PC6/I5

CAMERA(相机):
OV2640



- **Hardware Platform:**
Kiwi Nano 4K v1.2 Kit + OV2640 Sensor
- **Algorithms:**
Gaussian Filter | Median Filter | Hybrid Filter
- **Evaluation:**
Performance vs. Resource Utilization

TEAM: FPT_Bulbasaur

1. Trương Quốc Trung
2. Lê Quang Dũng

COMMON NOISE PHENOMENA & PROCESSING CHALLENGES

- Phenomenon 1: **Gaussian** noise (Random variation in brightness)
- Cause: **High temperature** during continuous operation → **Thermal Noise**
- Phenomenon 2: Vertical and horizontal **stripes** (Banding noise)
- Cause: 50Hz light frequency **conflicting** with Rolling Shutter timing



→ MCU Bottleneck: **SEQUENTIAL** PROCESSING (High Latency)



→ FPGA Solution: **PARALLEL** PROCESSING (Real-time)

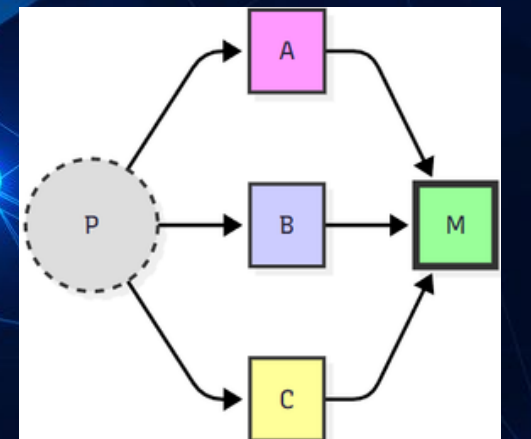


IMAGE NOISE AND GAUSSIAN FILTER

Formula: Gaussian Kernel
Total Weight = 16

1	2	1
2	4	2
1	2	1

Sum (Pixel × Kernel) = 452

→ **Normalized Pixel** = $452 \gg 4 = 28$

Input: 12 18 32 20 **50** 38 12 08 28

→ 1 2 1 2 **4** 2 1 2 1

12	20	12
18	50	08
32	38	28

Output: 12 18 32 20 **28** 38 12 08 28

12	20	12
18	28	08
32	38	28

12	20	12
18	23	08
32	38	28

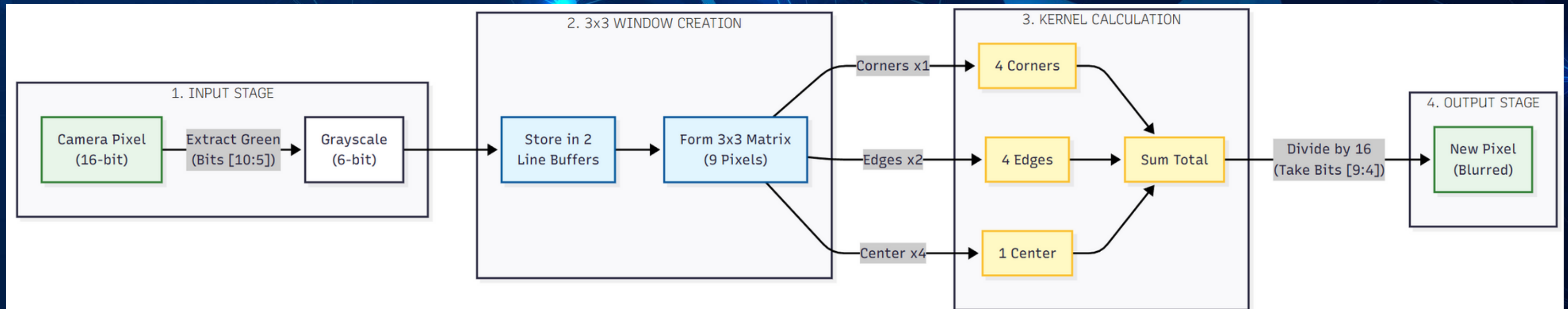
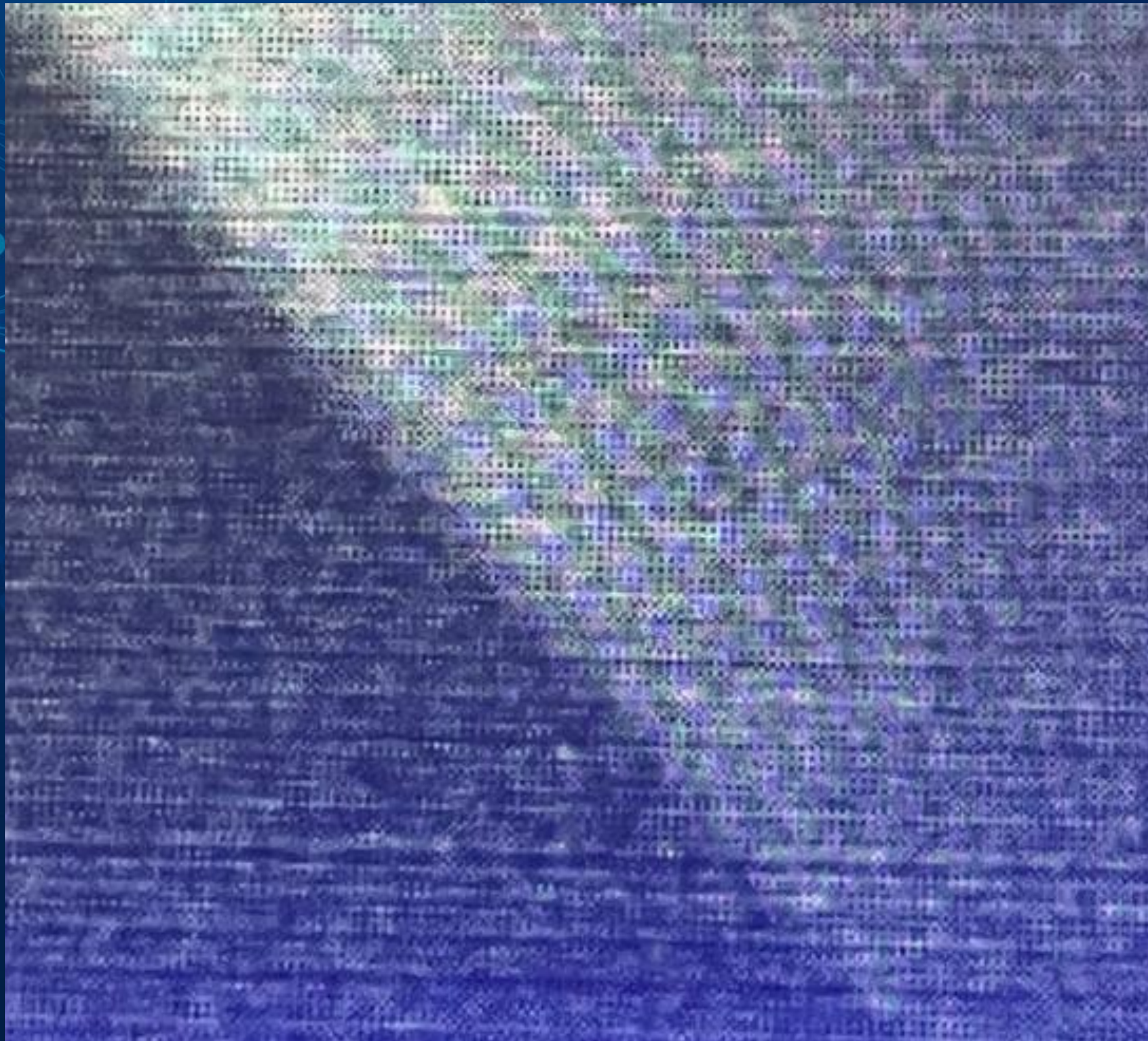


IMAGE NOISE AND GAUSSIAN FILTER

EXPERIMENTAL RESULTS

Before



After



EXPERIMENTAL
RESULTS

SALT-AND-PEPPER NOISE AND MEDIAN FILTER

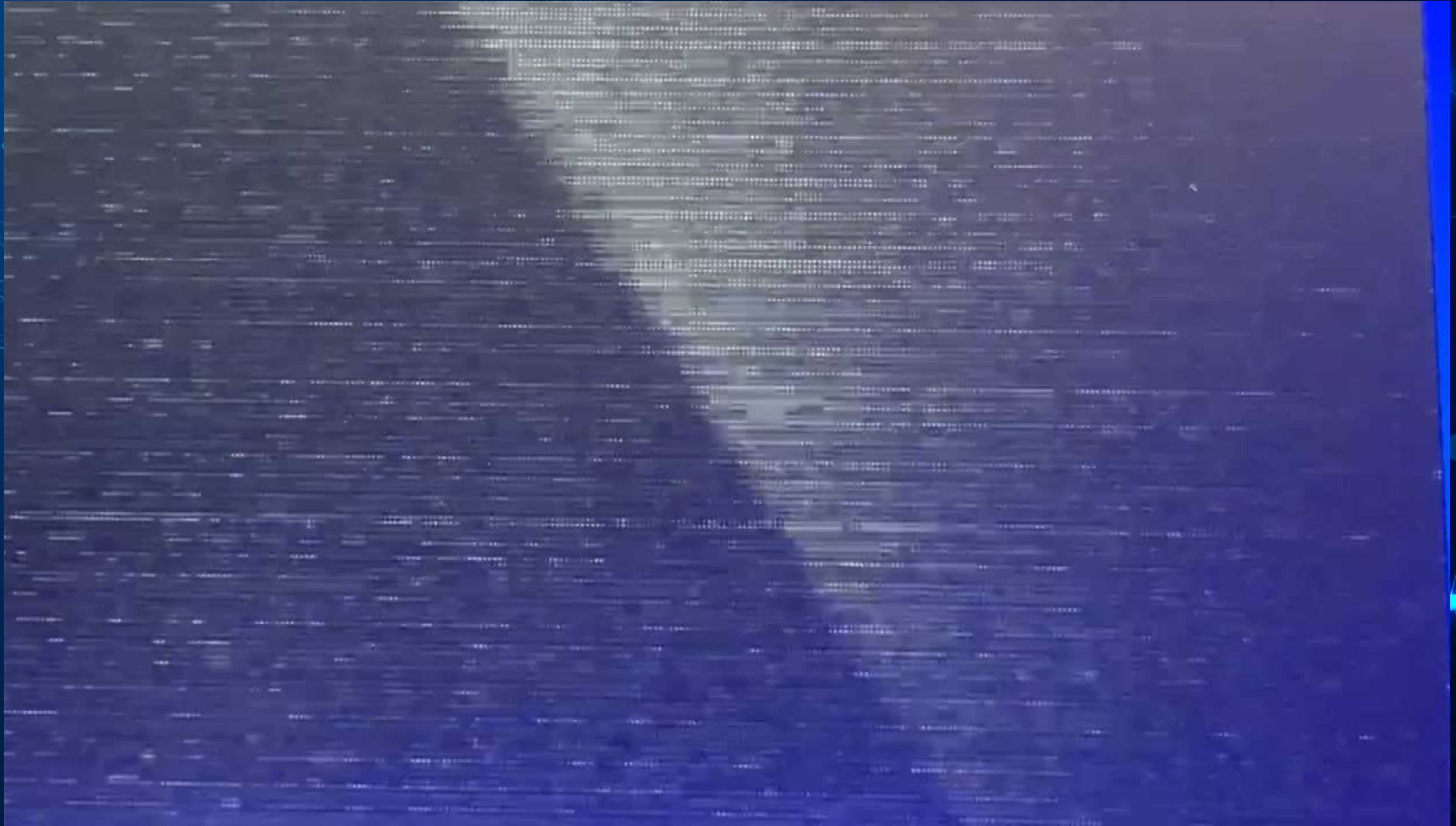
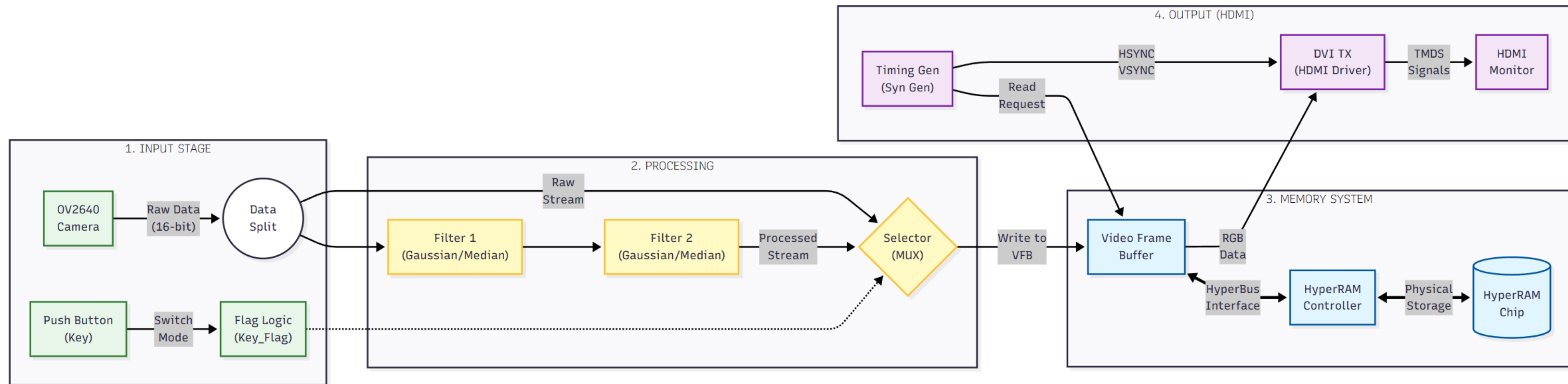


IMAGE NOISE AND GAUSSIAN FILTER

DATA FLOW



SALT-AND-PEPPER NOISE AND MEDIAN FILTER

Principle: Select the **center pixel** after **sorting** in ascending order.

Sorting: Ascending Order

[08, 12, 12, 18, **20**, 28, 32, 38, 50]

Median: The center value is **20**

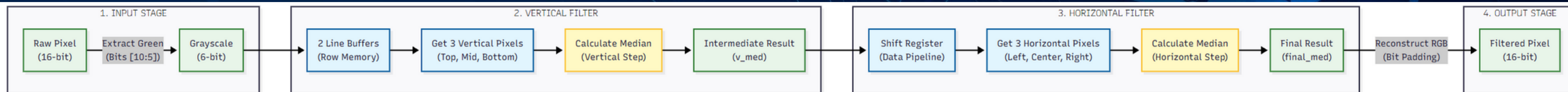
Input: 12 18 32 20 **50** 38 12 08 28

Output: 12 18 32 20 **20** 38 12 08 28

12	20	12
18	50	08
32	38	28

12	20	12
18	20	08
32	38	28

12	20	12
18	20	08
32	38	28



SALT-AND-PEPPER NOISE AND MEDIAN FILTER

EXPERIMENTAL RESULTS

Before



After



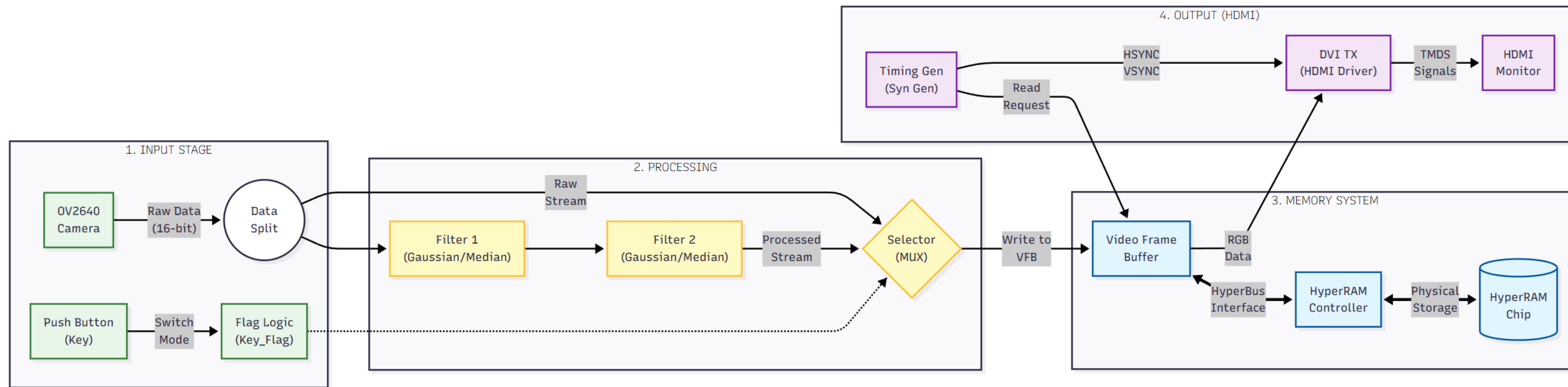
**EXPERIMENTAL
RESULTS**

IMAGE NOISE AND GAUSSIAN FILTER



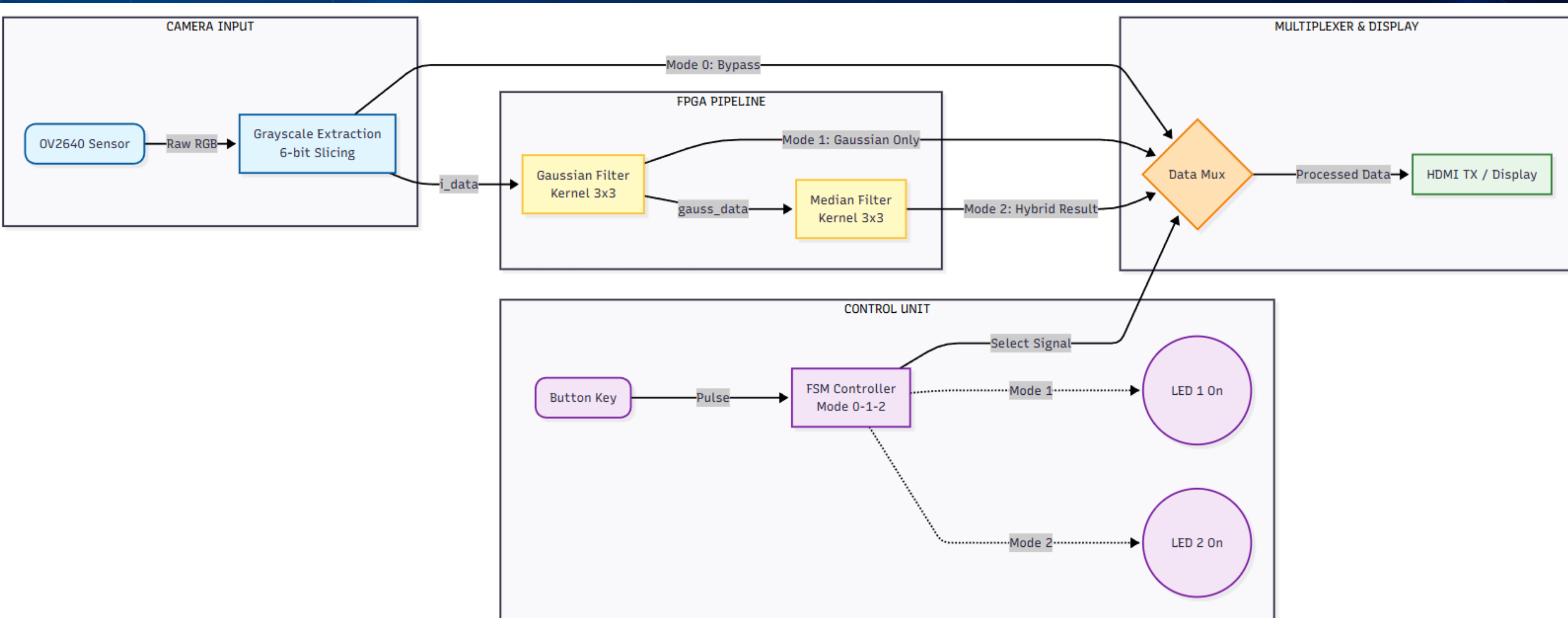
SALT-AND-PEPPER NOISE AND MEDIAN FILTER

DATA FLOW



SYSTEM ARCHITECTURE – HYBRID FILTER

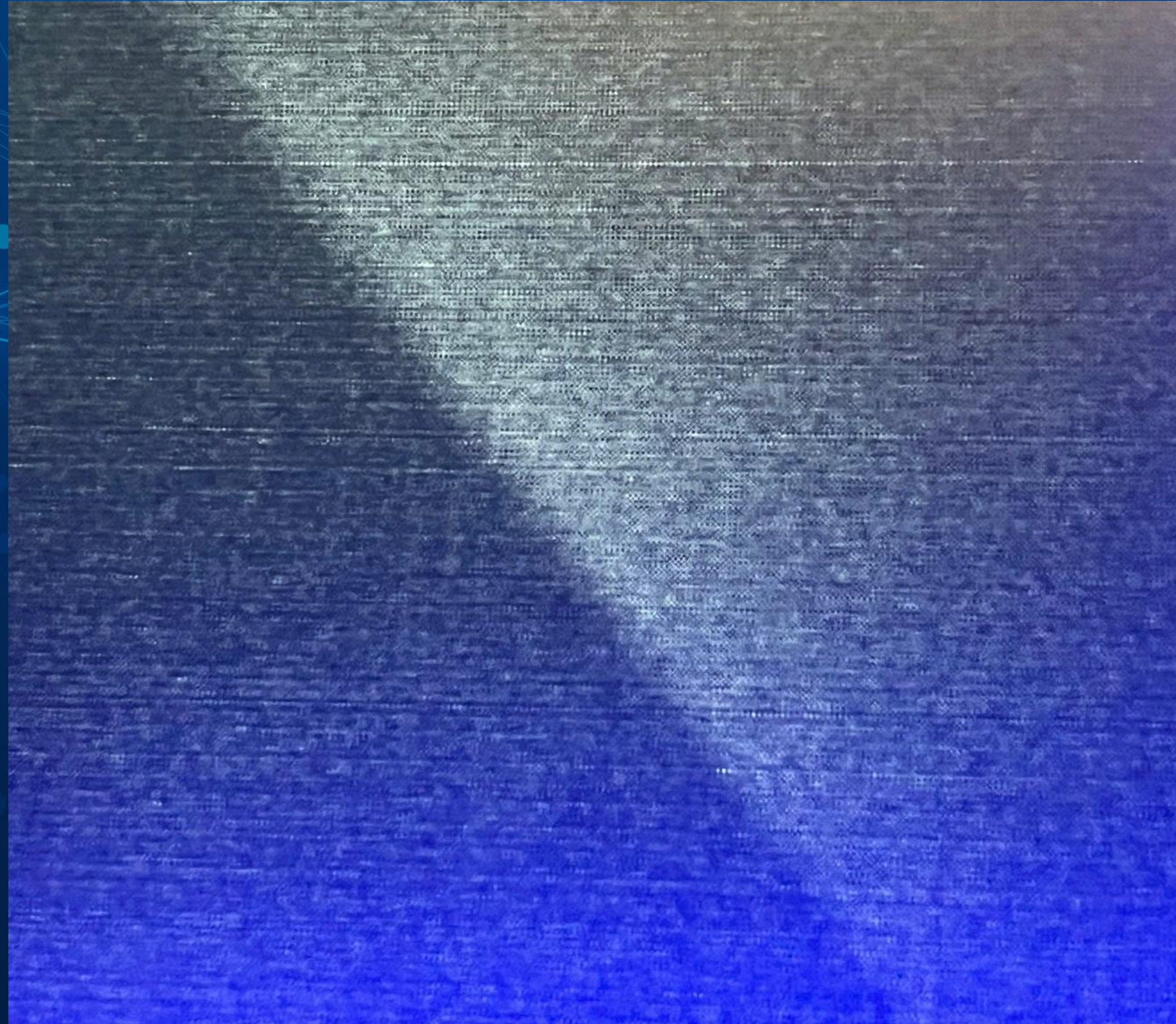
Pipeline Architecture: Camera → Gaussian → Median → HDMI → Display (Zero Latency)



SYSTEM ARCHITECTURE – HYBRID FILTER

EXPERIMENTAL RESULTS

Before



After





FPGA RESOURCE UTILIZATION SUMMARY

	Median	Normal	Gaussian	Synthesis Report
Resource Utilization Summary	Logic	UsageUtilization	UsageUtilization	UsageUtilization
	Register	1913(1523 LUT, 390 ALU) / 460842%	1957(1525 LUT, 432 ALU) / 460843%	1912(1636 LUT, 276 ALU) / 460842%
	--Register as Latch	1230 / 360935%	1183 / 360933%	1188 / 361233%
	--Register as FF	0 / 36090%	0 / 36090%	0 / 36120%
		1230 / 360935%	1183 / 360933%	1188 / 361233%
	BSRAM	10 / 10100%	10 / 10100%	6 / 1060%
Clock Summary	Clock Name	TypePeriodFrequency(MHz)RiseFall	TypePeriodFrequency(MHz)RiseFall	TypePeriodFrequency(MHz)RiseFall
	I_clk	Base37.03727.00.00018.519	Base37.03727.00.00018.519	Base37.03727.0000.00018.519
	PIXCLK	Base20.00050.00.00010.000	Base20.00050.00.00010.000	Base20.00050.0000.00010.000
	GW_PLLVR_inst/pllvr_inst/CLKOUT.default_gen_clk	Generated6.289159.00.0003.145	Generated6.289159.00.0003.145	Generated6.289159.0000.0003.145
	GW_PLLVR_inst/pllvr_inst/CLKOUTP.default_gen_clk	Generated6.289159.00.0003.145	Generated6.289159.00.0003.145	Generated6.289159.0000.0003.145
	GW_PLLVR_inst/pllvr_inst/CLKOUTD.default_gen_clk	Generated12.57979.50.0006.289	Generated12.57979.50.0006.289	Generated12.57979.5000.0006.289
	GW_PLLVR_inst/pllvr_inst/CLKOUTD3.default_gen_clk	Generated18.86853.00.0009.434	Generated18.86853.00.0009.434	Generated18.86853.0000.0009.434
	TMDS_PLLVR_inst/pllvr_inst/CLKOUT.default_gen_clk	Generated2.694371.30.0001.347	Generated2.694371.30.0001.347	Generated2.694371.2500.0001.347
	TMDS_PLLVR_inst/pllvr_inst/CLKOUTP.default_gen_clk	Generated2.694371.30.0001.347	Generated2.694371.30.0001.347	Generated2.694371.2500.0001.347
	TMDS_PLLVR_inst/pllvr_inst/CLKOUTD.default_gen_clk	Generated80.80812.40.00040.404	Generated80.80812.40.00040.404	Generated80.80812.3750.00040.404
	TMDS_PLLVR_inst/pllvr_inst/CLKOUTD3.default_gen_clk	Generated8.081123.80.0004.040	Generated8.081123.80.0004.040	Generated8.081123.7500.0004.040
	u_clkdiv/CLKOUT.default_gen_clk	Generated13.46874.20.0006.734	Generated13.46874.20.0006.734	Generated13.46874.2500.0006.734
	HyperRAM_Memory_Interface_Top_inst/u_hpram_top/clkdiv/CLKOUT.default_gen_clk	Generated12.57979.50.0006.289	Generated12.57979.50.0006.289	Generated12.57979.5000.0006.289
	Max Frequency Summary	Clock Name	ConstraintActual FmaxLogic Level	ConstraintActual FmaxLogic Level
I_clk		27.0(MHz)130.6(MHz)5	27.0(MHz)131.8(MHz)5	27.000(MHz)99.460(MHz)5
PIXCLK		50.0(MHz)73.0(MHz)13	50.0(MHz)100.8(MHz)8	50.000(MHz)78.117(MHz)8
TMDS_PLLVR_inst/pllvr_inst/CLKOUTD.default_gen_clk		12.4(MHz)142.2(MHz)5	12.4(MHz)142.2(MHz)5	12.375(MHz)78.253(MHz)5
u_clkdiv/CLKOUT.default_gen_clk		74.2(MHz)61.7(MHz)12	74.2(MHz)61.7(MHz)12	74.250(MHz)46.526(MHz)12
HyperRAM_Memory_Interface_Top_inst/u_hpram_top/clkdiv/CLKOUT.default_gen_clk		79.5(MHz)85.8(MHz)9	79.5(MHz)85.8(MHz)9	79.500(MHz)66.607(MHz)9

PERFORMANCE COMPARISON

Normal	Gaussian	Median	Hybrid
